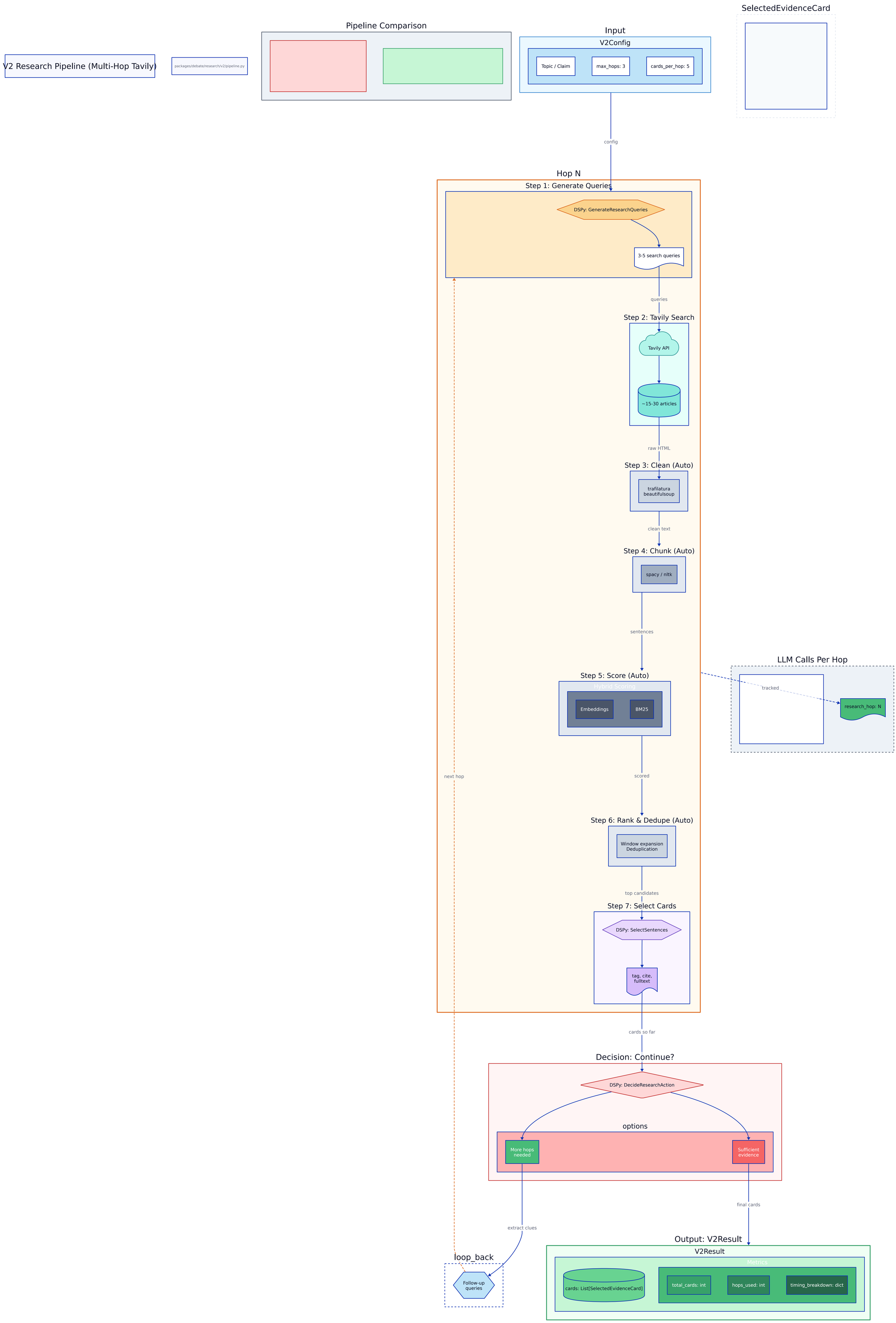
V2 Research Pipeline (Multi-Hop Tavily)
packages/debate/research/v2/pipeline.py
Pipeline Comparison
Input
V2Config
Topic / Claim
max_hops: 3
cards_per_hop: 5
SelectedEvidenceCard
config
Hop N
Step 1: Generate Queries
DSPy: GenerateResearchQueries
3-5 search queries
queries
Step 2: Tavily Search
Tavily API
~15-30 articles
raw HTML
Step 3: Clean (Auto)
trafilatura
beautifulsoup
clean text
Step 4: Chunk (Auto)
spacy / nltk
sentences
Step 5: Score (Auto)
Hybrid Scoring
Embeddings
BM25
LLM Calls Per Hop
tracked
research_hop: N
next hop
scored
Step 6: Rank & Dedupe (Auto)
Window expansion
Deduplication
top candidates
Step 7: Select Cards
DSPy: SelectSentences
tag, cite,
fulltext
cards so far
Decision: Continue?
DSPy: DecideResearchAction
options
More hops
needed
Sufficient
evidence
final cards
extract clues
loop_back
Follow-up
queries
Output: V2Result
V2Result
Metrics
cards: List[SelectedEvidenceCard]
total_cards: int
hops_used: int
timing_breakdown: dict